Chenyuan Zhou

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Education

Shanghai Jiao Tong University 2023.9 - 2027.6(expected)
B.S in computer science, **ACM Honors Class**

GPA(Core Courses): **3.94**/4.3
Selected courses:

- Algorithm Design and Analysis :**A+, 98/100**
- Compiler Design and Implementation:**A+, 99/100**
- Computer Architecture:**A+, 97/100**
- Data Structure:**A+, 97/100**
- Mathematical Analysis(Honor):**A+, 95/100**

Research interest

My research interests are Computer Vision, Multimodal Learning, and Continual Learning. I am especially interested in building multimodal systems that can perceive 3D structure, follow instructions, and continually adapt from new experiences in the real world.

Research Experience

Stanford CS, Stanford University 2026.6 - now

Research intern conducting experiments and participating in research on multimodal learning and continual learning, co-advised by Yejin Choi and Leonidas Guibas.

RHOS (Yong-Lu Li, Ce-Wu Lu) 2025.6 - 2026.6

Worked on precise manipulation with visual instruction and diverse tool adaptation under the supervision of Yong-Lu Li and Ce-Wu Lu.

Projects

Partial Rollout for LLM RL (LLM Course Project) Spring 2025 Link to Repo

Decomposed the single rollout process into multiple turns and propagated unfinished rollouts to the next iteration, reducing the negative impact of long-tail rollouts.
- Achieved a 30% increase in overall speed without performance degradation.
- Implemented partial code and conducted experiments in collaboration with Zeng Ji and Li Zhiyan. Details in PR 1826

RISCV CPU (Course Project) Fall 2024 Link to Repo

Designed a CPU in Verilog implementing the basic RISCV instruction set.
- Implemented branch prediction and instruction prefetching.
- Successfully executed on FPGA.

Mx Compiler (Course Project) Summer 2024 Link to Repo

A compiler for Mx* (an educational language with basic features of C), as well as Clang (with `mem2reg` and register allocation).
- Implemented various optimizations including SCCP, DCE, inlining, GVN & GCM, loop detection, and unrolling.
- Utilized SSA graph coloring to allocate registers based on liveness analysis.

Papers

HIOTA: Human-Inspired One-Shot Tool Affordance Adaptation
(in submission to CoRL 2026)

Developed HIOTA, a human-inspired tool affordance learning framework that captures transferable tool-use priors and enables one-shot adaptation to unseen tools, as a co-leader.

- Modeled tool affordances from unguided exploration, tool observation, and task-directed demonstrations that reveal how tools should be used to accomplish goals.
- Grounded visual and geometric cues with abstract manipulation patterns into an affordance latent space, improving real-robot adaptation to novel tools while preserving learned skills.

Awards and Honors

2023, 2024 Zhiyuan Honors Scholarship (**2 %** in SJTU)

Student Work and Teaching Experience

Computer Programming, Teaching Assistant	2024 Fall
Data Structure, Teaching Assistant	2025 Spring
Principle and Practice of Computer Algorithms(AI), Teaching Assistant	2025 Summer
Vice Monitor of ACM Honors Class 2024	2024.9 - 2025.9

Skills

Programming	C++, Python, Java, Verilog,
Tools	Git, Solidworkers, ros2, LaTeX
Languages	Chinese(Native speaker), English(Fluent)

Last updated: June 29, 2026